Studying the Effects of Temperature and Artifacts on Video Moment Retrieval with Small Language Models

BACHELOR THESIS PROJECT

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INTRODUCTION

Video Moment Localization (VML) finds precise temporal segments in a video matching a natural language query. Text-only methods fail on vague or abstract queries. This work proposes a multimodal pipeline fusing text and reference images into enriched query representations, and systematically investigates two factors: sampling temperature and artifact-prompt type.

PROBLEM STATEMENT

Given video V and query Q_i, predict timestamps $(\hat{t}_s, \hat{t}_e)$ of the matching moment.
Multimodal query is formed as:

$$Q_{\text{multi}} = \text{Summarize}(Q_i,\ I_i^{(m)})$$

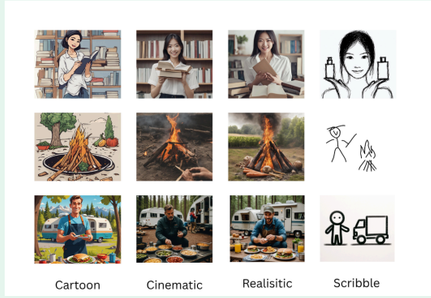
where $I_i^{(m)}$ is a reference image under modality $m \in \{\text{cartoon, cinematic, realistic, scribble}\}$.
Localization objective:

$$(\hat{t}_s,\ \hat{t}_e) = \arg\max_{t_s,\, t_e} \mathcal{S}(\phi(V_{t_s:t_e}),\ \psi(Q_{\text{multi}}))$$

$\phi(\cdot)$ — video segment embedding (HERO) $\quad \psi(\cdot)$ — multimodal query embedding $\quad \mathcal{S}(\cdot,\cdot)$ — semantic similarity
Temperature study: $\tau \in \{0.2, 0.4, 0.6, 0.8, 1.0\}$ with 3 prompt variants $\mathcal{T}^{(1)}, \mathcal{T}^{(2)}, \mathcal{T}^{(3)}$ per temperature.
Artifact study: fixed $\tau = 0.2$, ten prompt templates $\mathcal{T}^{(p)}$, $p \in \{0, \ldots, 9\}$.

LITERATURE REVIEW

- Early text-only VML methods struggled with vague queries, motivating multimodal setups like ICQ-Highlights, which pair each query with an image to reduce ambiguity and improve grounding.
- Multimodal LLMs such as Qwen3-VL-8B-Instruct, LLaVA-Mistral-7B, and InternVL2-8B fuse text and visual cues into compact, semantically rich query representations for better event understanding.
- Strong video encoders like HERO capture spatial details and long-range temporal structure, producing robust video features for retrieval.
- Transformer-based grounding models such as Moment-DETR use these features with multimodal query embeddings to directly predict precise temporal boundaries.

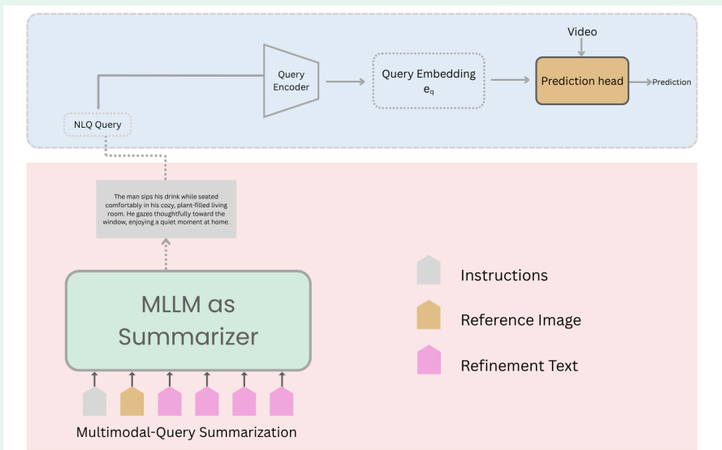


Dataset Overview

METHODOLOGY

Stage 1 — <u>Dataset Preprocessing:</u>
Extract NLQs, four image modalities, and refinement text from ICQ-Highlights, normalize all fields, and prepare clean inputs.

Stage 2 — <u>Multimodal Summarization:</u>
Use instruction-guided prompts with Qwen/LLaVA to generate ≤2-sentence summaries for each modality, followed by strict post-processing.



Architecture

Stage 3 — <u>Feature Extraction:</u>
Convert each summary into a query embedding compatible with Moment-DETR and generate frame/clip-level video representations using the HERO transformer encoder.

Stage 4 — <u>Moment Retrieval:</u>
Combine summary embeddings with HERO video features inside Moment-DETR, where the decoder uses cross-attention to predict precise start–end timestamps.

Stage 5 — <u>Evaluation:</u>
Measure localization quality using IoU overlap, Recall@1 at 0.5/0.7 thresholds for top predictions, and mAP to assess retrieval consistency across IoU levels.

EXPERIMENTAL SETUP

1 — Hardware & Environment
Experiments ran on an RTX 3090 (24 GB) using Python 3.10, PyTorch 2.3.1, and FP16 inference for all three models, with all seeds fixed for reproducibility.
2 — Dataset & Summarization
ICQ-Highlights was used with four image modalities: Cartoon, Cinematic, Realistic, and Scribble. Each modality (~1546 samples) was summarized independently using Qwen3-VL-8B-Instruct, LLaVA-Mistral-7B, and InternVL2-8B, with preprocessing, instruction prompts, and strict two-sentence output control.
3 — Retrieval Pipeline
Summaries were encoded and combined with HERO video features, then fed into Moment-DETR, which predicted temporal boundaries. Performance was evaluated per modality using R1@IoU=0.5, R1@IoU=0.7, and mAP averaged over IoU thresholds 0.3, 0.5, and 0.7.
4 — Temperature Sensitivity Study
General prompts p∈{0,1,2} evaluated at τ∈{0.2,0.4,0.6,0.8,1.0} across all three models and four modalities— 180 total runs.
5 — Artifact-Type Study
Ten prompt templates covering ten visual-change categories evaluated at fixedτ=0.2 across all models and modalities — 120 total runs.

RESULTS

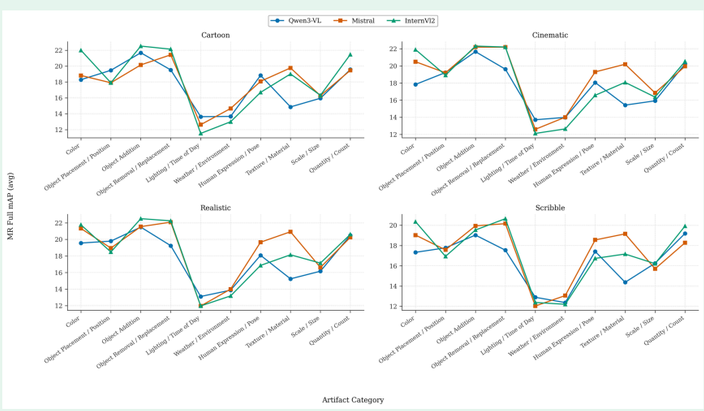
Model–Modality	T=0.2	T=0.4	T=0.6	T=0.8	T=1.0
Qwen-Cartoon	22.95±0.35	22.57±0.25	23.00±0.23	**23.02±0.49**	22.67±0.19
Qwen-Cinematic	**23.04±0.48**	22.93±0.61	23.03±0.48	22.79±0.33	23.12±0.37
Qwen-Realistic	23.02±0.57	**23.65±0.74**	23.27±0.25	23.49±0.52	22.91±0.10
Qwen-Scribble	20.90±0.38	21.08±0.61	20.98±0.54	**21.13±0.71**	20.71±0.52
Mistral-Cartoon	**22.54±0.85**	21.81±0.43	21.73±0.56	21.19±0.16	20.66±0.28
Mistral-Cinematic	**23.29±0.81**	23.04±0.56	22.71±0.09	22.81±0.29	22.03±0.55
Mistral-Realistic	**23.10±0.78**	22.95±0.74	22.14±0.23	21.93±0.46	22.01±0.61
Mistral-Scribble	**20.52±0.64**	20.38±0.16	20.04±0.69	19.75±0.46	19.41±0.35
InternVL2-Cartoon	**23.04±0.27**	22.91±0.36	22.85±0.30	22.44±0.29	22.02±0.09
InternVL2-Cinematic	23.52±0.15	23.53±0.23	**23.66±0.29**	23.01±0.19	22.80±0.09
InternVL2-Realistic	**23.78±0.04**	23.76±0.30	23.08±0.29	23.27±0.36	22.91±0.09
InternVL2-Scribble	**22.05±0.51**	21.72±0.29	21.34±0.21	21.05±0.25	20.35±0.46

(a) mAP

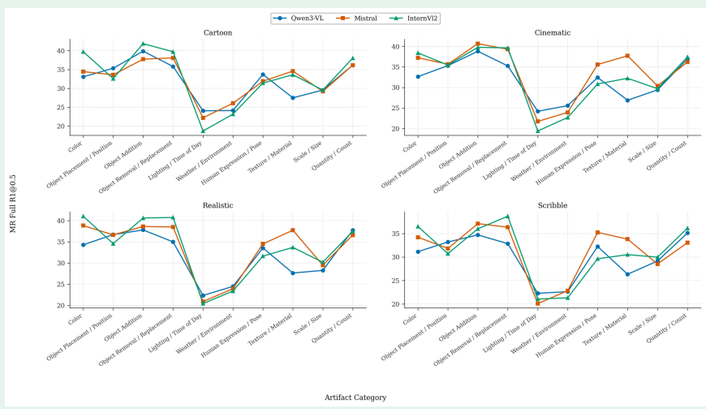
Model–Modality	T=0.2	T=0.4	T=0.6	T=0.8	T=1.0
Qwen-Cartoon	41.77±0.96	41.29±0.88	41.65±0.51	**42.09±1.04**	41.29±0.99
Qwen-Cinematic	41.36±0.89	**41.89±1.18**	41.54±1.01	41.25±0.32	41.10±0.42
Qwen-Realistic	42.25±0.83	42.43±1.06	**42.57±1.07**	42.55±0.40	40.95±0.49
Qwen-Scribble	**38.45±0.71**	38.41±0.82	38.59±0.50	38.75±1.01	38.34±0.87
Mistral-Cartoon	**41.02±1.93**	39.07±0.87	39.53±0.92	38.59±0.66	37.01±0.27
Mistral-Cinematic	**41.95±1.01**	41.68±1.54	41.54±0.49	40.49±0.93	39.30±1.13
Mistral-Realistic	**40.79±1.22**	40.92±0.80	39.71±0.90	39.37±1.12	39.48±0.33
Mistral-Scribble	**37.47±0.98**	36.40±0.12	36.83±1.21	35.74±1.22	35.71±0.56
InternVL2-Cartoon	**41.50±0.58**	41.25±0.16	40.51±0.47	40.06±0.19	39.69±0.34
InternVL2-Cinematic	41.86±0.58	**42.32±0.31**	42.16±0.74	41.11±0.13	40.92±0.82
InternVL2-Realistic	**42.55±0.25**	**42.55±0.23**	41.95±0.94	42.21±0.77	39.87±0.25
InternVL2-Scribble	**39.73±0.12**	39.25±0.67	38.77±0.12	38.75±0.98	36.58±0.86

(b) R1@IoU = 0.5

Table 1 - Temperature sensitivity across metrics (mean±std).



(a) mAP



(b) R1@IoU = 0.5

Table 2 - Artifact Sensitivity across metrics

- Low temperature (τ=0.2) gives best and most stable results across all models.
- Object-level artifact prompts boost retrieval; Lighting and Weather prompts hurt it by 8–10 mAP points.
- InternVL2-8B with general prompt at τ=0.2 is the best overall configuration (mAP = 23.10).
- General prompts consistently outperform all artifact-specific prompts by 1.0–1.5 mAP points across all models and modalities.

REFERENCES

- L. Li, Y.-C. Chen, Y. Cheng, Z. Gan, L. Yu, and J. Liu, "Hero: Hierarchical encoder for video+language omni-representation pre-training," 2020.
- G. Zhang, M. L. A. Fok, J. Ma, Y. Xia, D. Cremers, P. Torr, V. Tresp, and J. Gu, "Localizing events in videos with multimodal queries," 2024
- J. Lei, T. L. Berg, and M. Bansal, "Qvhighlights: Detecting moments and highlights in videos via natural language queries," 2021.